

TSA SERIES

MODEL

TSA700-6N

The TSA700-6N is a horizontal-multi-joint SCARA robot built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. 6 kg payload, 700 mm arm reach, 200 mm Z stroke and a fast 0.44 s standard cycle, in a compact 24 kg body. Integrated pneumatic and signal harness through the arm keeps end-effector tooling clean and repeatable.

4-Axis SCARA	Repeatability	Texsonics RC	Site-ready
Payload 6 kg · Reach 700 mm	±0.02 mm (XY) positional accuracy	In-house controller, drives and teach pendant	Floor / Wall / Ceiling mount, factory calibrated

01

SPECIFICATIONS

MECHANICAL		ENVIRONMENT / CONTROL	
AXES	4 (SCARA)	SIGNALS	15 pin D-Sub (13-return cable)
PAYLOAD	6 kg	PNEUMATICS	1 × □6 + 2 × □6 (0.59 MPa)
ARM LENGTH	700 mm	HARNESS LENGTH	3 m
Z STROKE	200 mm	PROTECTION	IP20
STANDARD CYCLE	0.44 s	ENVIRONMENT	0–40 °C, 35–85 % RH
BODY WEIGHT	24 kg		

02 OVERVIEW

HIGH-THROUGHPUT PLANAR MOTION, 0.44 S CYCLE.

The TSA700-6N is a 4-axis SCARA built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. Integrated pneumatic and signal harness keeps end-effector tooling clean and repeatable.

01**FAST CYCLE**

0.44 s standard cycle, 8700 mm/s XY speed and 2000°/s tool rotation for the shortest possible takt times.

02**HARNESS-THROUGH ARM**

15-pin D-Sub signal and dual 6 mm pneumatic lines routed through the arm keep tooling wiring clean.

03**PLANAR ACCURACY**

± 0.02 mm XY, ± 0.01 mm Z, $\pm 0.01^\circ$ tool — precision suited to assembly and inspection cells.

03 PRODUCT

TSA700-6N • TSA SERIES

04 WORKING RANGE

MOTION — GROUP A

A-AXIS (INNER ARM)	±140°
B-AXIS (OUTER ARM)	±145°
Z-AXIS (VERTICAL STROKE)	200 mm
W-AXIS (TOOL ROTATION)	±360°
A/B MAX SPEED	8700 mm/s

MOTION — GROUP B

Z MAX SPEED	1100 mm/s
W MAX SPEED	2000°/s
REPEATABILITY — A/B	±0.02 mm
REPEATABILITY — Z	±0.01 mm
REPEATABILITY — W	±0.01°

05 CONTROL & INTEGRATION

CONTROLLER

Texsonics RC series
in-house motion controller,
dual-kernel real-time OS

TEACH PENDANT

11.5" touch tablet
drag-teaching, Wi-Fi,
on-device offline
programming

FIELDBUS

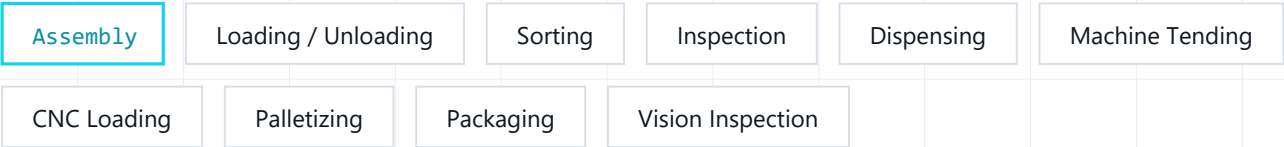
**EtherCAT / Modbus
TCP**
industrial fieldbus,
dual-protocol support

SOFTWARE

TEXCAM offline CAM
toolpath generation, digital
twin simulation, vision

06 APPLICATIONS

Deployable across the primary applications the TSA700-6N was engineered for — full production-line integration with the Texsonics control stack.



07 BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

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TSA700-6N

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TSA SERIES
4-AXIS SCARA · PAYLOAD 6 kg · REACH 700 mm

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